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CS C26 09 EU 001

NETCORE SYSTEM LTD NETWORK TOPOLOGY

Below is a full set up of the company network with three department with the server in the IT department.

The network Topology was setup for NETCORE SYSTEM LTD using two computers for each department connected to a switch with these switches connected to Router using a copper straight-through copper. The server was connected to the IT Department.

The IT server was configured with static IP address to allow consistent address all the services and Default Gateway with Subnet Mask was set

HTTP (hypertext transfer Protocol)service was turn on

A username and password was created on the FTP server and this was tested and response was successful. Configuration of the router interface was done the command line interface for all the three Department with their IP address and Subnet Masks

The three interface were turn on

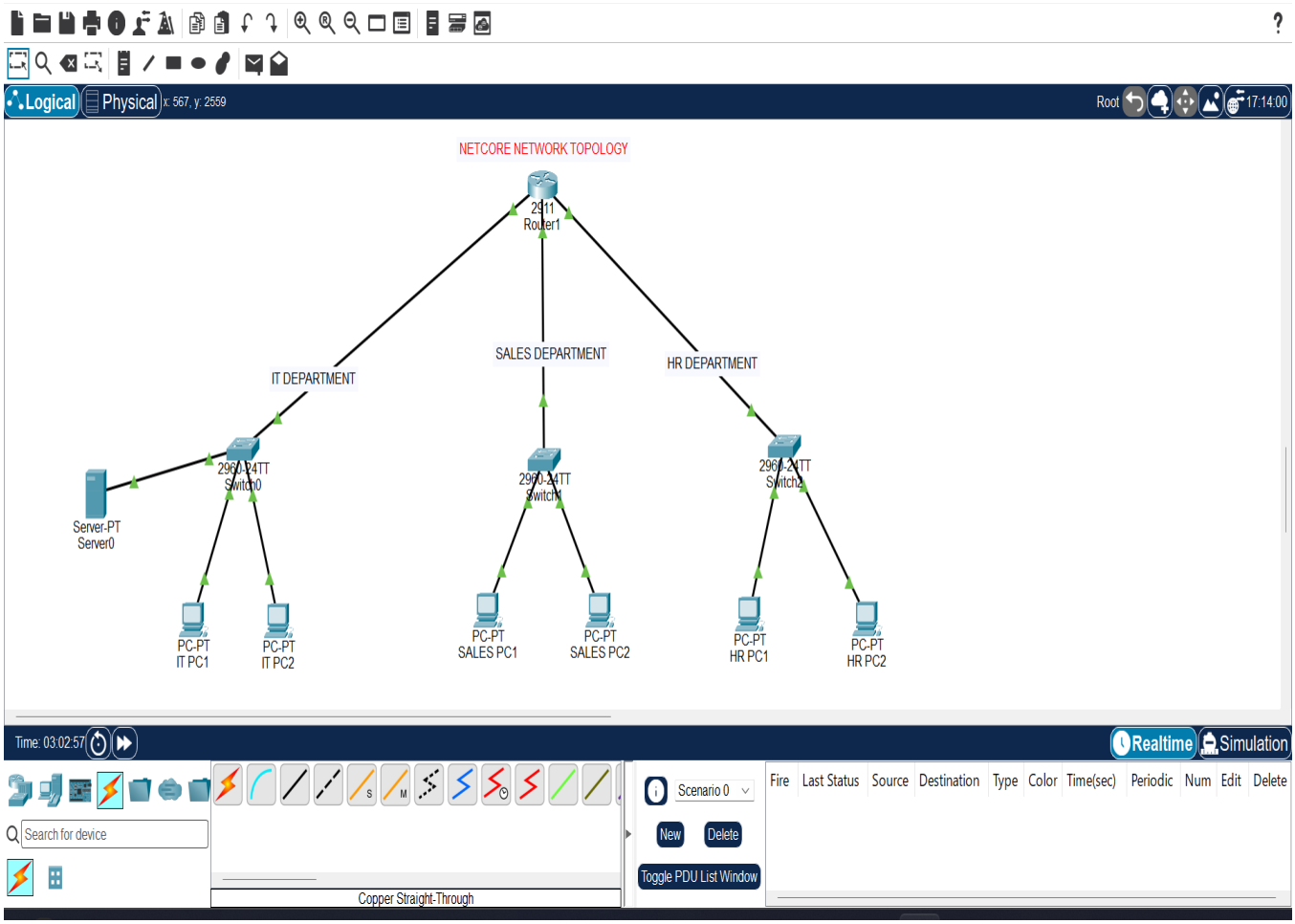
The router was able to provide address for all the six PCs

This was verify with Desktop via Command prompt ipconfig and it was successful

A test of Web Browser was also successful

Since Netcore Systems Ltd didn't want the Sale PCs to connectivity to IT PCs, therefore the ACL was use block ICMP Echo traffic from Sales to IT

This was done through the Router Command Line



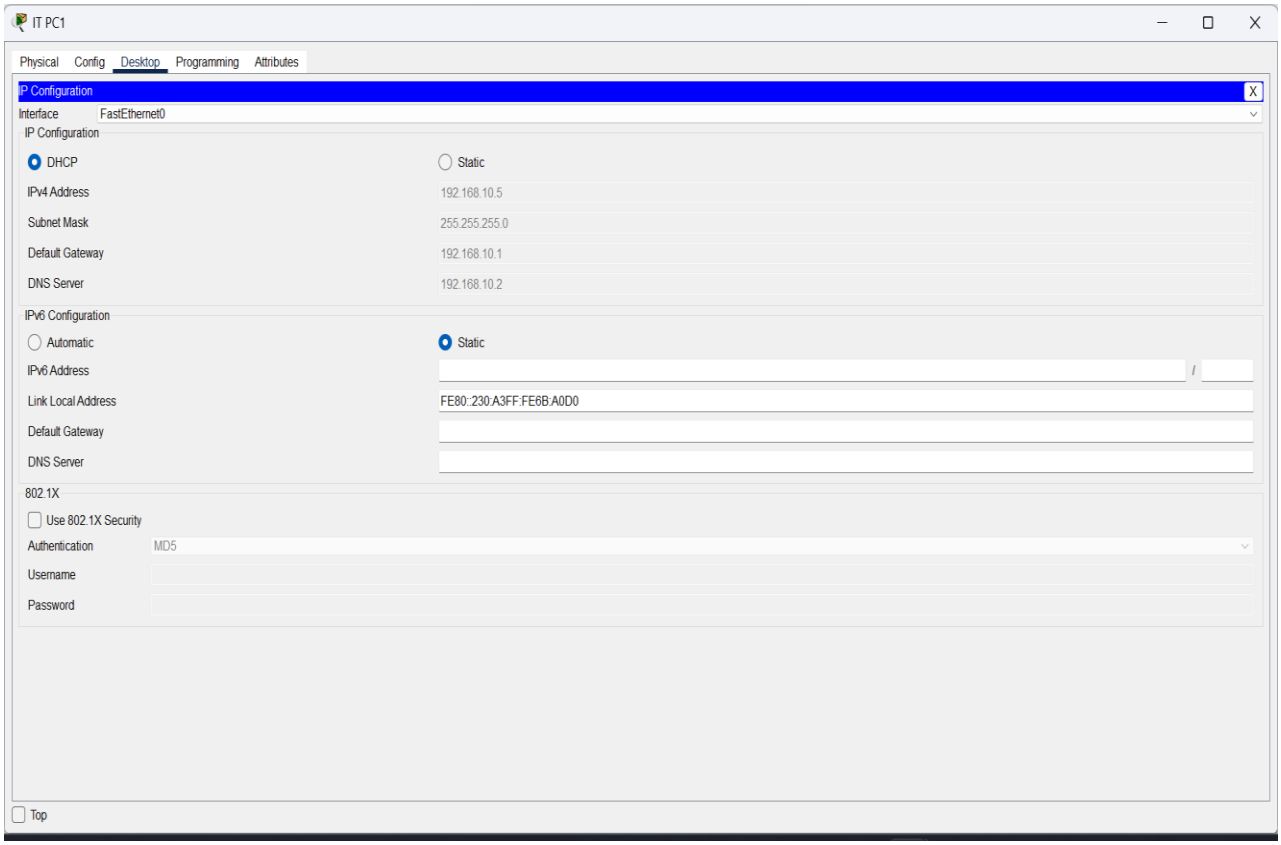
The IP address was assigned to every computer using DHCP server and configure all router interface with correct subnet addresses

Because Server in the IT Department was assigned with static IP address as well as turning the DHCP server on

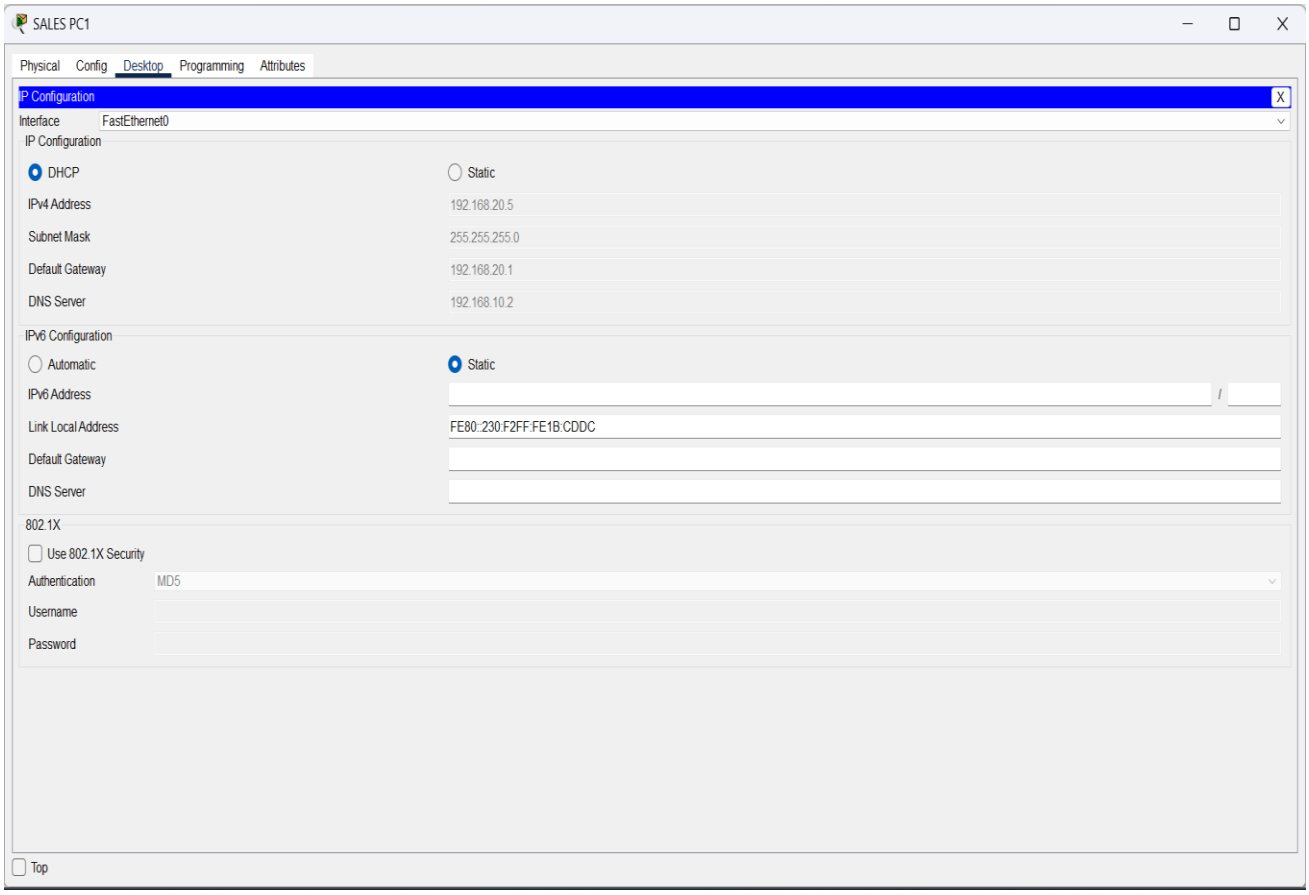
Using the DHCP all the PC in all the department were assigned with IP addresses

A range of IP address were assigned to each department

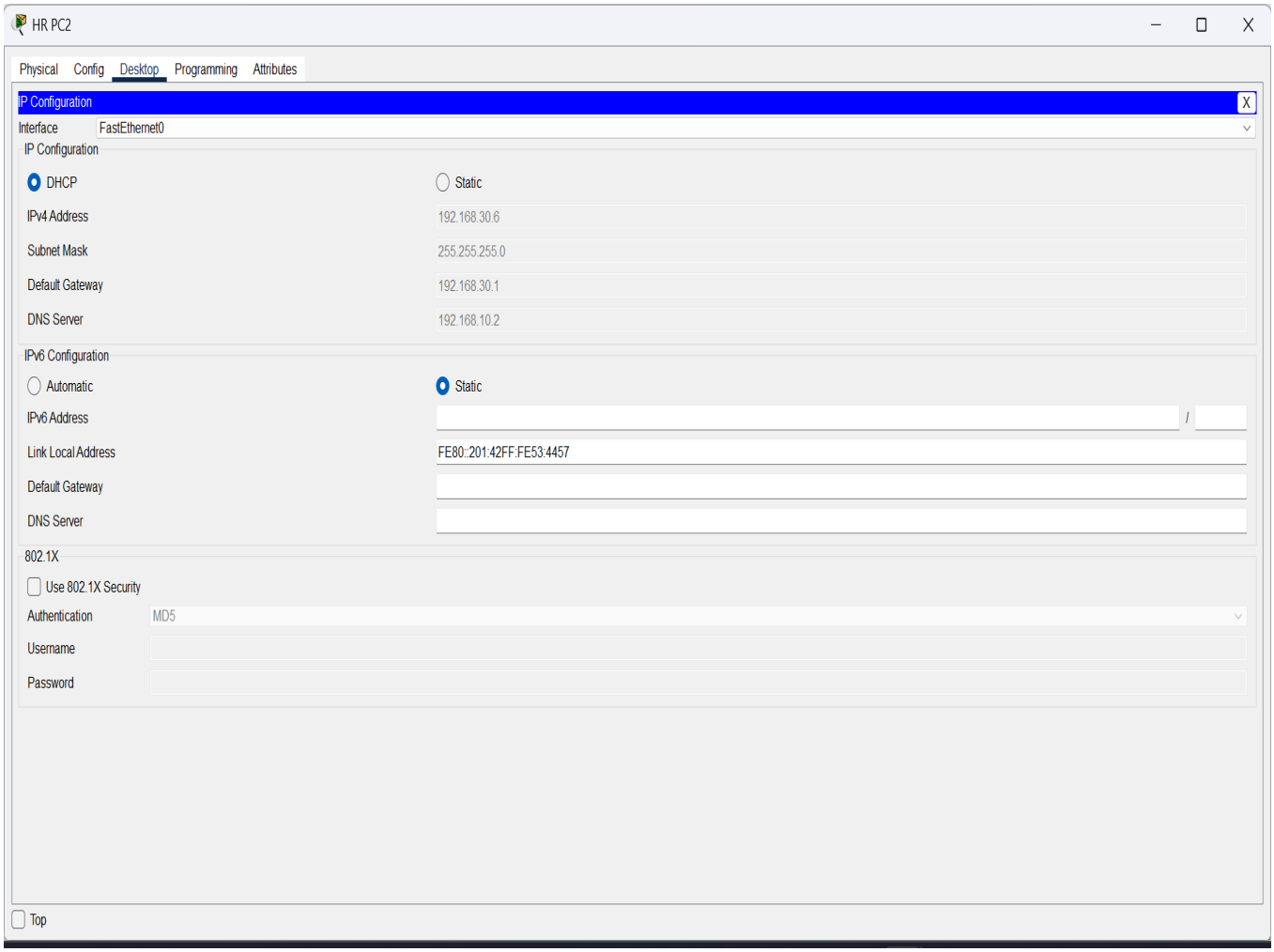
Screenshot of IP address configuration on IT PC using DHCP Server



Screenshot of IP address configuration on SALES PC using DHCP Server

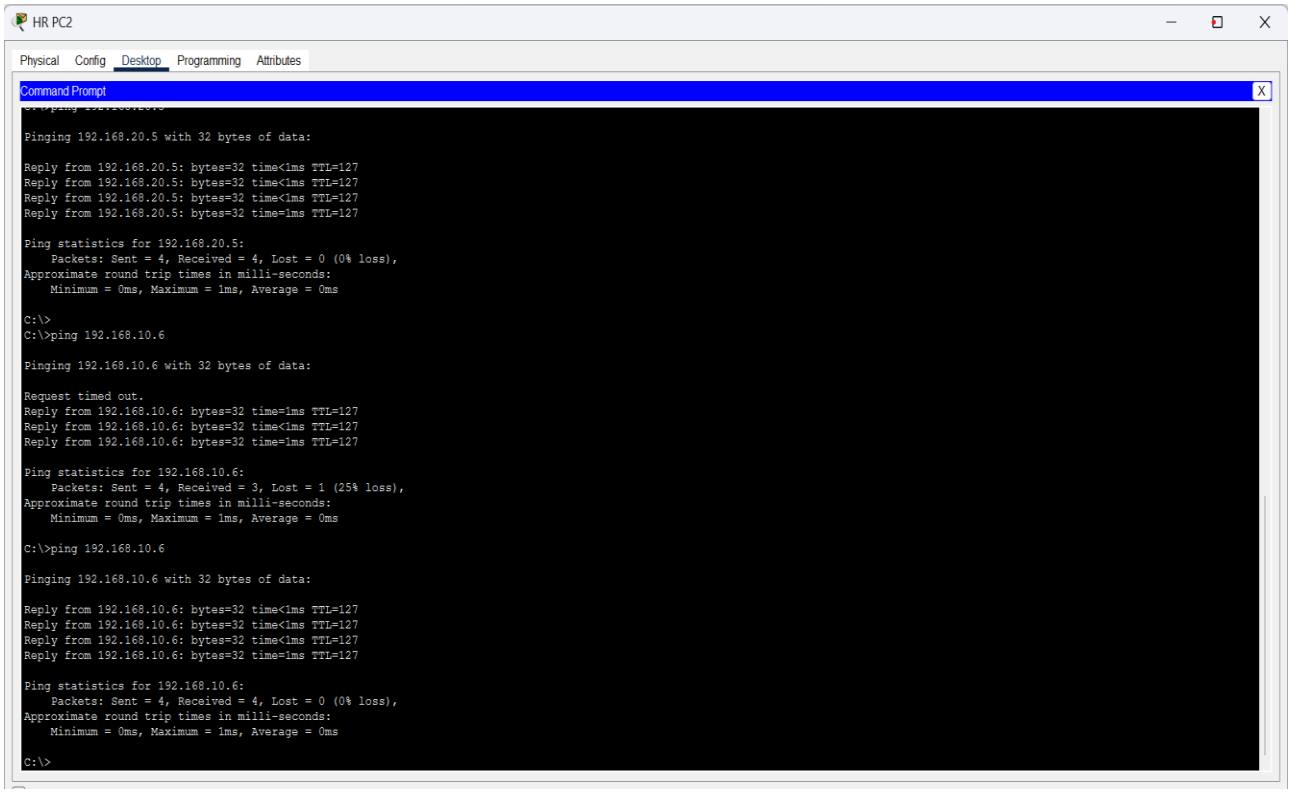


Screenshot of IP address configuration on HR PC using DHCP Server



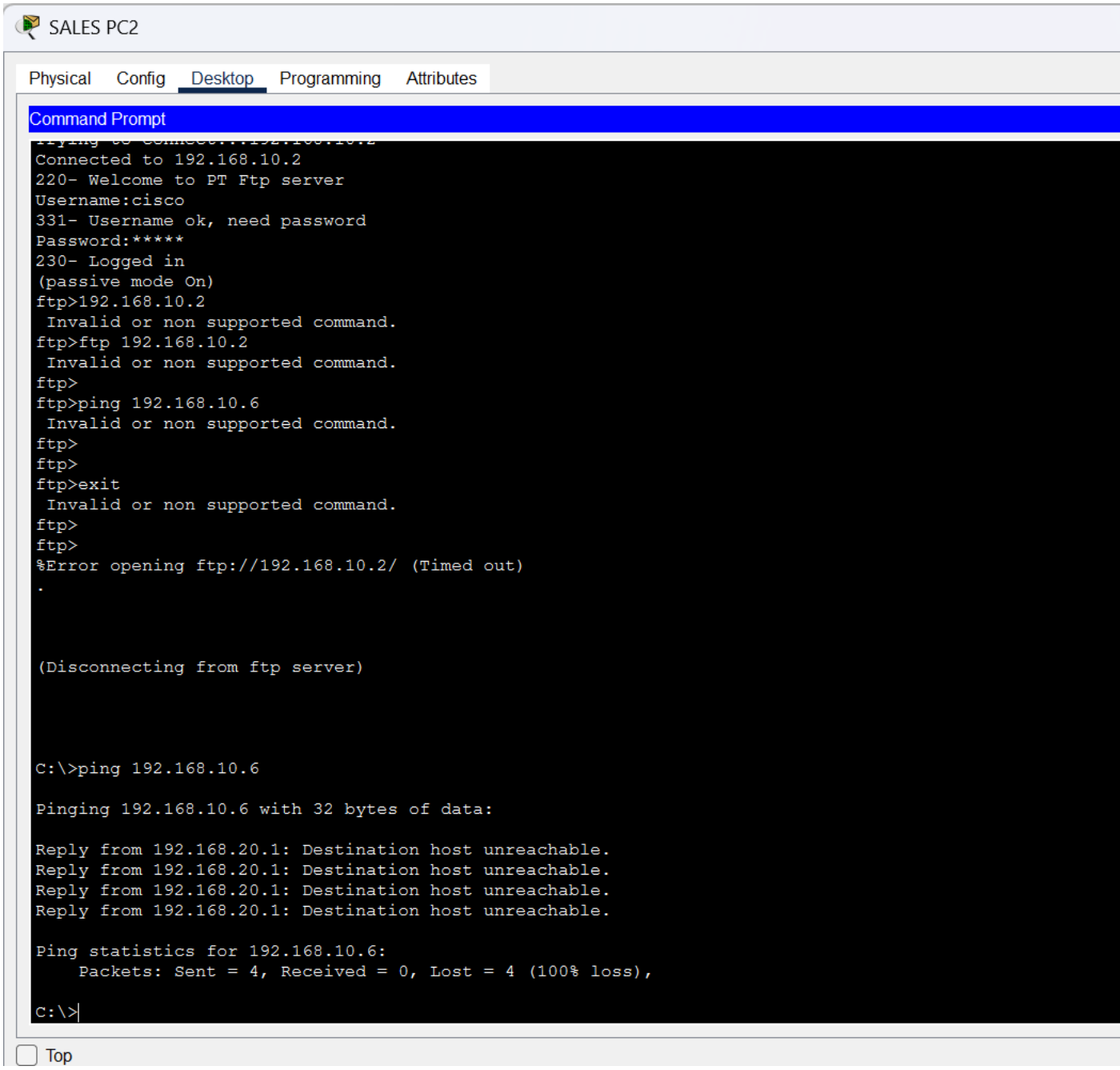
Screenshot of successful ping from HR to IT PC

Using the Command prompt this is the screen shot of basic connectivity between HR and IT pcs



Screenshot of fail ping from a Sales PC (PC2) to IT PC

Using the command prompt below is the failed connectivity between Sales to IT because of the applied ACL



Screenshot of ACL command as entered in the router CLI

Router1

Physical Config CLI Attributes

IOS Command Line Interfac

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Press RETURN to get started.



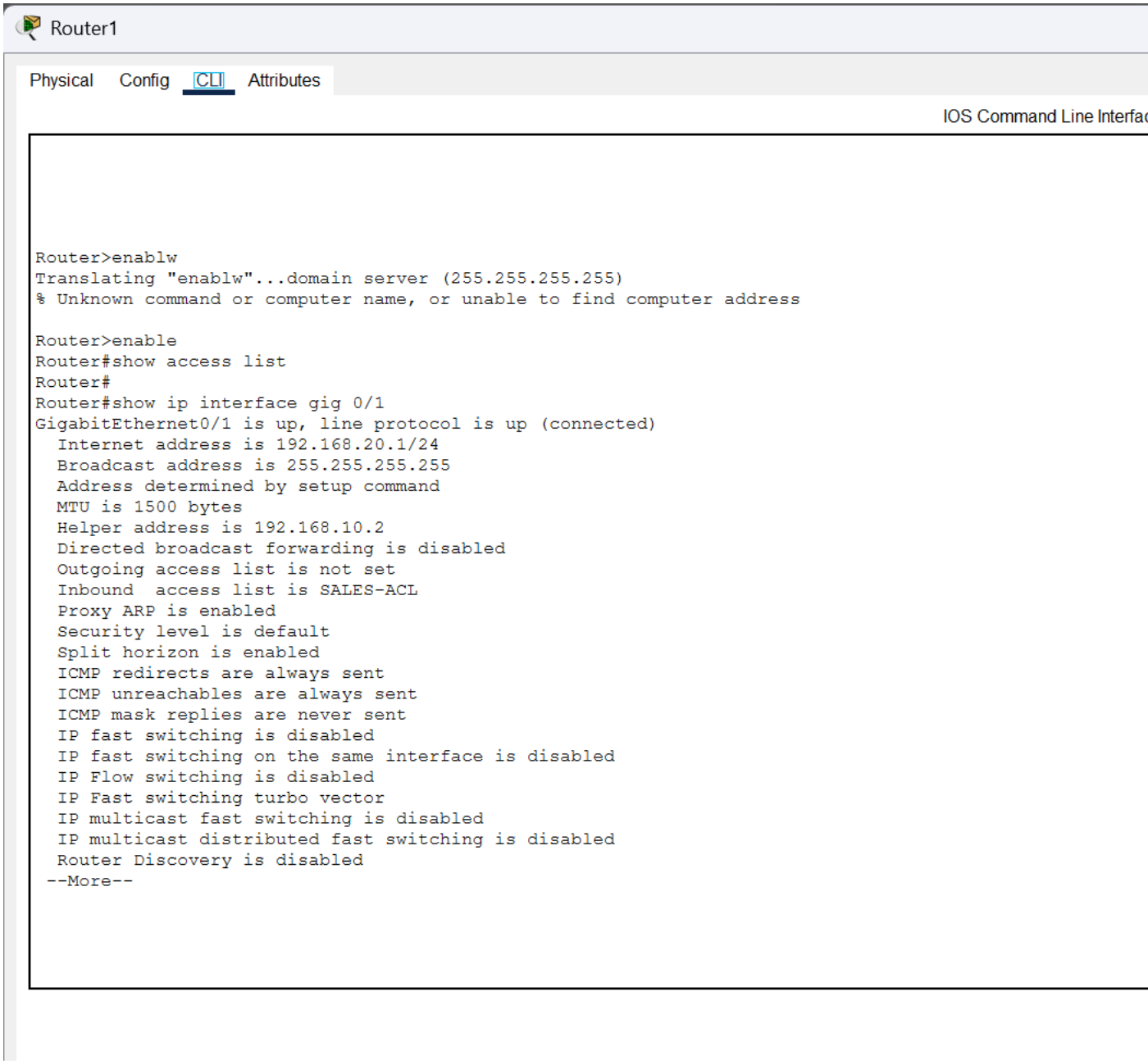

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended SALES-ACL
Router(config-ext-nacl)#deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#interface gig 0/1
Router(config-if)#ip access-qroup SALES-ACL in
                              ^
% Invalid input detected at '^' marker.

Router(config-if)#ip access-group SALES-ACL in
Router(config-if)#exit
Router(config)#



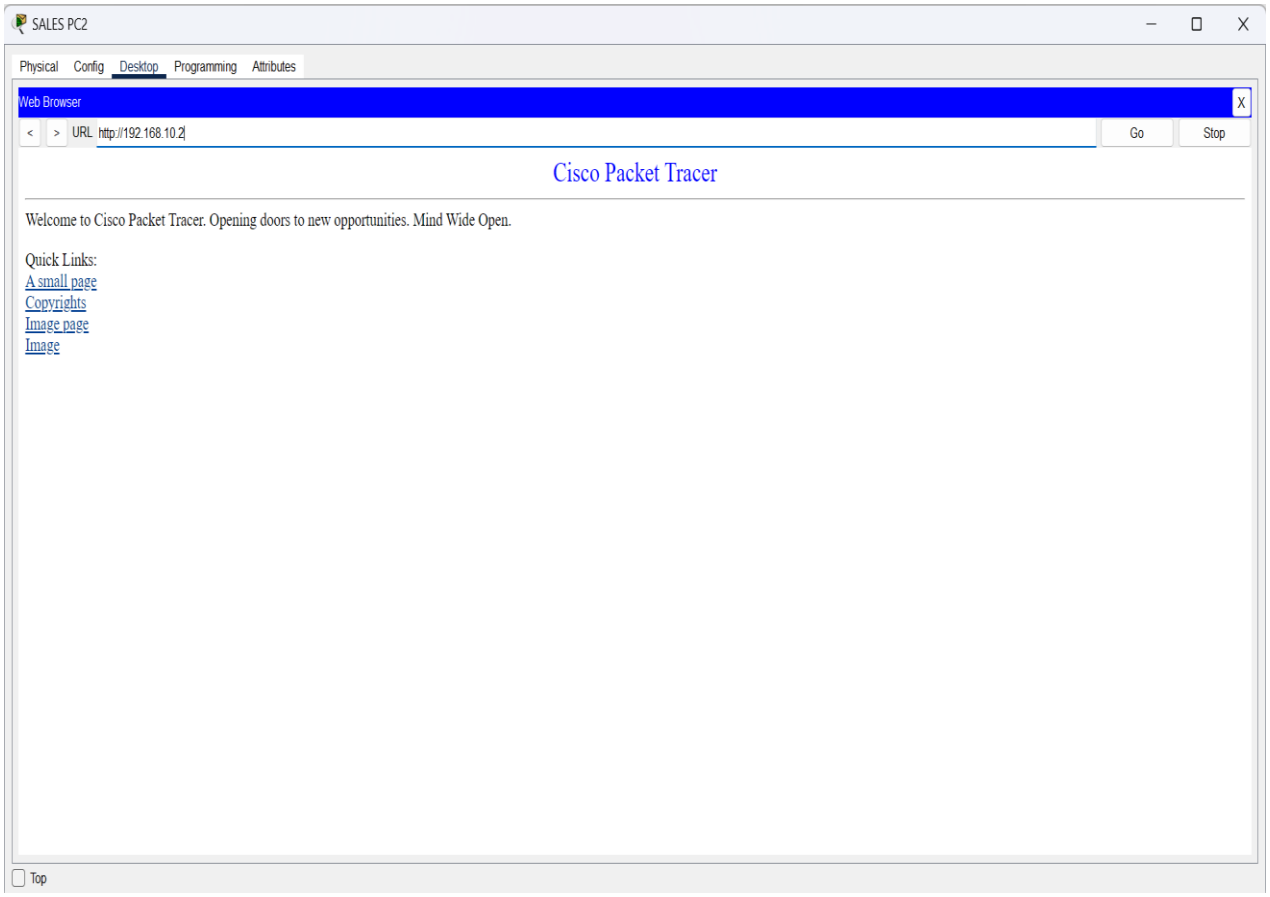

Router con0 is now available
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Top



Screenshot of SALES having access to the http service of the server

Even though there is no connectivity from Sale to IT below is the evidence of sale been able to access the webserver.


SALES PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
URL
http://192.168.10.2
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image
Top